

AI/ML Capstone Project
PR1 to PR7 Weekly Progress Report (5% of final mark)
Instructor: Prof. Mouhamed Abdulla

Name 1: Afolabi Adesina	Student Number 1: 991808102
Name 2: Evans Frimpong	Student Number 2: 991771602
Name 3: Xian Qin	Student Number 3: 991578381
Name 4: Yanan Yang	Student Number 4: 991841037

Q1: Summary of Work

Provide a summary of the work completed during this current sprint. [1 mark]

- **Yanan:** Trained three machine learning models, achieving accuracies around 82%. However, the F-1 scores were found to be extremely low for Class 2 (serious) and Class 3 (fatal). After conducting an investigation, it was confirmed that the dataset is heavily imbalanced: Class 1 (slight) accounts for 82.7%, while Class 2 and Class 3 account for only 15.6% and 1.6% respectively. This skew reflects real-life traffic accident distributions but poses a significant training challenge.
- **Evans:** Reviewed the application frontend framework and identified the critical API connections required to establish full functionality between the client side and the backend.
- **Xian:** Successfully test-ran Python scripts to fetch environmental variables via APIs from a government weather website. Additionally, discovered and reviewed a new dataset containing road condition images.
- **Afolabi:**
 - **Ethics & Fairness:** Investigated fairness metrics for the project to mitigate systemic bias in data collection.
 - **Literature Review:** Integrated two core research papers into the project framework: Paper 1 focusing on weather-aware trajectory planning and the SPI formula, and Paper 2 covering ML traffic prediction benchmarks.
 - **EDA & Schema Audit:** Explored both the City of Toronto and UK Department for Transport (DfT) datasets, conducted complete schema audits, and plotted feature distributions.
 - **Summary Statistics:** Replicated Paper 2's weather, road surface, and lighting factor analysis on local DfT data using custom bar charts and correlation heatmaps.
 - **Safety Score Design:** Calibrated individual weights for the E_{index} formulation using empirical paper findings (e.g., snow +47%, dusk +22%, standing water +21%).
 - **Preprocessing:** Handled missing values, engineered key temporal and environmental features (hour of day, day of week, is night, season), and executed label encoding.
 - **Correlation Analysis:** Computed Pearson correlation matrices, Chi-Square tests of independence, and Point-Biserial coefficients.
 - **Feature Selection:** Developed a three-method voting pipeline (combining Chi-Square, Mutual Information, and Random Forest Gini importance) to filter and select 8 optimal features.

- **Model Training & Tuning:** Built a comprehensive model pipeline training 5 baseline models (Logistic Regression, Decision Tree, KNN, Random Forest, and LightGBM). Conducted hyperparameter optimization using GridSearchCV with StratifiedK-Fold cross-validation for Logistic Regression ($L1/L2$), Random Forest, and LightGBM.
- **Model Comparison Matrix:** Generated a comprehensive performance evaluation table measuring Accuracy, Precision, Recall, Macro/Weighted F1-score, Matthews Correlation Coefficient (MCC), and Area Under the ROC Curve (AUC).
- **Additional Data for Images:** Incorporated external image repositories for vision-based classification:
 - Extreme Road Image Dataset: [GitHub Repository](#)
 - RSCD-1million Road Surface Classification Dataset: [Hugging Face Dataset](#)

Q2: List of Challenges

Provide a list of challenges encountered, and how the team addressed those challenges during this sprint. [1 mark]

- **Yanan:** The extreme class imbalance (82.7% Class 1) means that a naive model predicting the majority class automatically achieves high accuracy while failing completely on minority classes. Attempting to pass `class_weight='balanced'` within Random Forest and Logistic Regression alongside one-hot encoding caused the models to aggressively over-predict Class 2 and Class 3. This indicates that the current feature space may not linearly or easily separate the classes, demanding deeper non-linear feature engineering.
- **Evans & Team:** Encountered structural difficulties in securing a comprehensive, high-quality, and fully annotated image dataset suitable for training a robust computer vision model for automated road condition checking.
- **Xian:** Encountered accessibility constraints and strict language/protocol barriers with certain official government weather APIs. This required adapting connection scripts and looking into alternative open data feeds to ensure stable real-time data flows.

Q3: Technology Stack

Provide the current technology stack used with specific version numbers for tools, libraries, databases, and APIs. [1 mark]

- **Yanan:** Google Colab, `pandas`, `scikit-learn` (`neural_network`, `linear_model`, `metrics`, `compose`, `pipeline`).
- **Evans:** Flask API, YOLOv8 (Ultralytics architecture for image detection).
- **Xian:** Python environment, Apache Packages, `pandas`, `opendatatoronto` API wrapper.
- **Afolabi:** VS Code, Jupyter Notebook, Cursor IDE, PyTorch, TensorFlow, LightGBM, and Git.

Q4: Meeting Minutes

Provide the details of the meeting minutes (those conducted in class and outside) and the GitHub activity. [1 mark]

- **Yanan:** Summarized the analytical findings from the current model iterations and proposed a multi-experimental framework to enhance minority class F-1 metrics.
- **Xian:** Formulated algorithmic optimization strategies to enhance computational efficiency during data ingestion.
- **Afolabi:** Suggested integrating real-time highway camera feeds to build a 3-way decision fusion model (combining Tabular, NLP, and Vision brains).
- **Evans:** Conducted a detailed architectural assessment of state-of-the-art image recognition networks to focus development efforts on high-yield models.

Q5: Future Work

Provide a work plan for the upcoming week. [1 mark]

- **Yanan:** Design and run advanced modeling experiments to address dataset imbalance and evaluate feature separation viability.
- **Evans:** Establish a standardized output pipeline for the image training system and formulate the exact mathematical logic to integrate vision metrics into the final road safety score.
- **Afolabi:** Utilize Cursor to manage the pipeline architecture. Build the NLP Brain via a TF-IDF extraction pipeline on Ontario 511 text alerts, construct the Vision Brain using OpenCV and a Custom CNN on Ontario road camera images, and create a centralized multi-modal dashboard fusing all three safety components.
- **Xian:** Merge and map the newly acquired road image datasets with the existing database schema, and continue engineering real-time data flow pipelines from public transit APIs.

References

- [1] J. Jiang, Y. Miao, and D. Wu, "Machine Learning based Prediction Analysis of Potential Factors in Traffic Accidents," *Applied and Computational Engineering*, vol. 99, no. 1, pp. 112–120, Nov. 2024. [Online]. Available: <https://doi.org/10.54254/2755-2721/99/20251788>
- [2] C. Palay, "SDOT collisions all years [Data set]," *City of Seattle ArcGIS Online*, 2023. [Online]. Available: <https://data-seattlecitygis.opendata.arcgis.com/>
- [3] Department for Transport, "Road safety data, collisions last 5 years [Data set]," 2025. [Online]. Available: <https://data.dft.gov.uk/road-accidents-safety-data/>
- [4] Microsoft, "Visual Studio Code Docs," 2024. [Online]. Available: <https://code.visualstudio.com/docs>
- [5] Project Jupyter, "Jupyter Notebook Documentation," 2024. [Online]. Available: <https://jupyter-notebook.readthedocs.io/>
- [6] Cursor, "Cursor Documentation," 2024. [Online]. Available: <https://cursor.com/docs>
- [7] PyTorch, "PyTorch Documentation," 2024. [Online]. Available: <https://pytorch.org/docs/>

- [8] TensorFlow, "TensorFlow API Documentation," 2024. [Online]. Available: https://www.tensorflow.org/api_docs
- [9] scikit-learn developers, "scikit-learn: Machine Learning in Python Documentation," 2024. [Online]. Available: <https://scikit-learn.org/stable/>
- [10] The pandas development team, "pandas Documentation," 2024. [Online]. Available: <https://pandas.pydata.org/docs/>
- [11] Pallets Projects, "Flask Documentation," 2024. [Online]. Available: <https://flask.palletsprojects.com/>
- [12] Ultralytics, "YOLOv8 Documentation," 2024. [Online]. Available: <https://docs.ultralytics.com/>
- [13] City of Toronto, "opendatatoronto: Access the City of Toronto Open Data Portal," 2024. [Online]. Available: <https://github.com/sharlagelfand/opendatatoronto>